



Rajshahi University of Engineering & Technology

Department of Electrical and Computer Engineering (ECE)

Lab Report - 02

Experiment No. 02

Observation of the Speed Control of a DC Shunt Motor by using
Armature Current Control Method and Field Current Control
Method

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Experiment No. 02

Experiment Title

Observation of the speed control of a DC shunt motor by using Armature Current Control Method and Field Current Control Method.

Objectives

- To understand the fundamental methods of speed control of a DC shunt motor.
- To observe the variation of motor speed with armature current under the Armature Current Control Method.
- To observe the variation of motor speed with field current under the Field Current Control Method.
- To plot and analyze the characteristic curves for both control methods and compare their performance.

Theory

The speed of a DC shunt motor is governed by the well-known EMF equation:

$$E_b = V - I_a R_a \quad (1)$$

and the back EMF is also expressed as:

$$E_b = \frac{P\phi ZN}{60A} \quad (2)$$

Combining the two relations, the speed of the motor can be expressed as:

$$N = \frac{V - I_a R_a}{\phi} \times \frac{60A}{PZ} \Rightarrow N \propto \frac{V - I_a R_a}{\phi} \quad (3)$$

where,

- V = Applied armature voltage (V)
- I_a = Armature current (A)

- R_a = Armature resistance (Ω)
- ϕ = Flux per pole (Wb)
- N = Speed (rpm)
- P = Number of poles, Z = Number of armature conductors, A = Number of parallel paths

From this equation, it is evident that the speed of a DC shunt motor can be controlled in two principal ways:

1. Armature Current Control Method

In this method, a variable resistor (R_{AR}) is connected in series with the armature circuit while the field current (and hence flux ϕ) is kept constant by keeping the field resistance fixed. As the armature resistance R_{AR} is decreased, the voltage drop across it decreases, and since $V - I_a R_a$ increases, the back EMF E_b increases, and consequently the speed increases. In practice, as the series resistance is decreased, more voltage is available across the armature, allowing the armature current to increase, which in turn raises the developed torque and speed. This method gives speed control below and up to the rated speed, but it is inefficient due to power loss ($I_a^2 R_{AR}$) in the series resistor.

2. Field Current Control Method

In this method, a variable resistor (R_F) is connected in series with the shunt field winding, while the armature voltage is kept constant. Increasing R_F decreases the field current I_f , which reduces the flux ϕ . Since $N \propto 1/\phi$ (for constant V and $I_a R_a$), a decrease in flux results in an increase in speed. This method is highly efficient since the field circuit carries a very small current, so the power loss in the rheostat is small. It provides speed control above the rated (base) speed and gives a wide range of smooth speed variation.

Comparison

- Armature control method gives speed below rated speed, is less efficient, and torque remains nearly constant (constant torque drive).
- Field control method gives speed above rated speed, is more efficient, and provides nearly constant power operation.

Required Apparatus

Table 1: Required Apparatus for Experiment No. 02

SI. No.	Apparatus	Ratings
1	Variable Resistor	Resistance: 5000 Ω , Current: 0.31 A
2	Variable Resistor	Resistance: 200 Ω , Current: 1.58 A
3	Fixed DC Power Supply	Voltage: 220 V
4	DC Multimeters	Voltage: 600 V, Current: 20 A
5	Three Phase Asynchronous Motor	Power: 500 W, Voltage: 400 V-Y/230 V- Δ , Current: 1.5 A-Y/2.6 A- Δ , Speed: 2850 rpm
6	DC Motor	Power: 300 W, Voltage: 220 V, Speed: 2500 rpm
7	Tachogenerator	Current: 0.07 A max, Speed: 5000 rpm max

Circuit Diagram

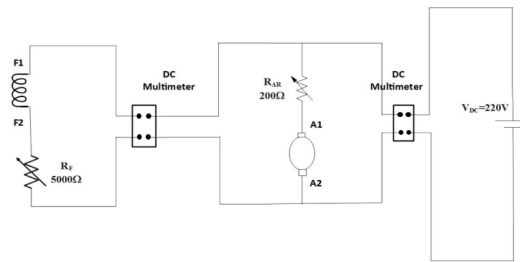
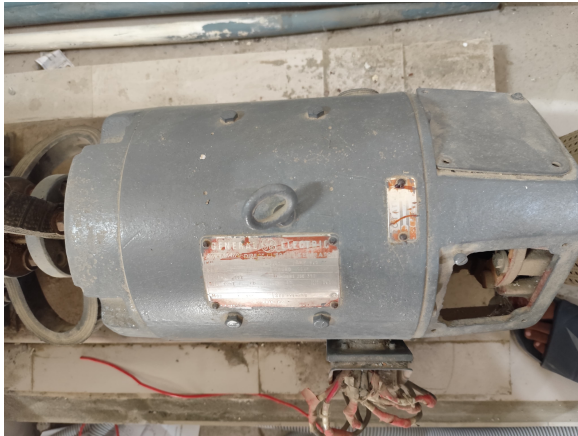


Figure 1: Required circuit diagram for speed control of DC shunt motor

The DC field winding (F_1 – F_2) is connected through a field rheostat R_F (5000 Ω) and a DC multimeter to measure field current I_f . The armature circuit (A_1 – A_2), which is coupled with a tachogenerator to measure speed, is connected through an armature rheostat R_{AR} (200 Ω) and a second DC multimeter to a fixed 220 V DC supply.

Experimental Setup

The photographs below show the actual laboratory apparatus used to perform this experiment.

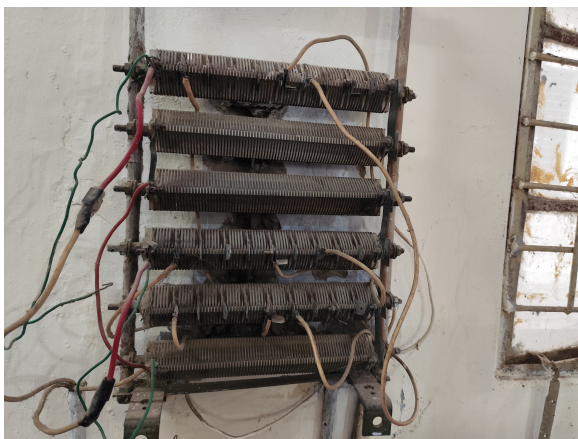


(a) Nameplate of the DC machine (dynamic direct current generator/motor)



(b) Nameplate of the three-phase asynchronous (induction) motor

Figure 2: Nameplate details of the coupled machine set



(a) Field rheostat (R_F) resistor bank

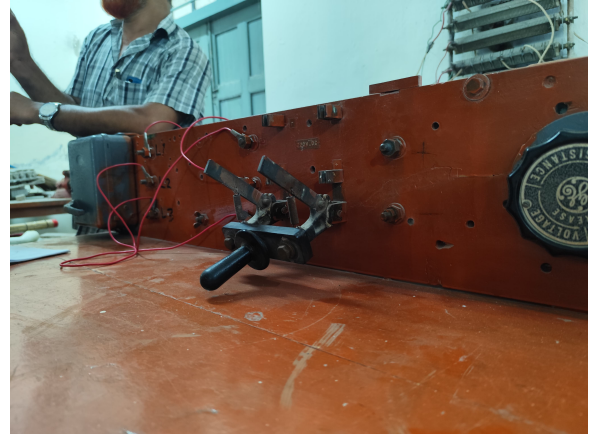


(b) Armature rheostat / voltage control unit

Figure 3: Variable resistors used for armature and field control



(a) DC motor coupled with the three-phase induction motor set



(b) Three-point starter panel used to start the DC motor

Figure 4: Coupled machine set and starter panel



Figure 5: Rotary control knob used to vary the rheostat setting

Procedure

1. The circuit was connected as shown in Figure 1.
2. The field rheostat R_F was kept at its maximum value and the armature rheostat R_{AR} was kept at its maximum value before switching ON the supply, to ensure a safe starting condition.
3. The DC supply was switched ON and the motor was started gradually using the 3-point starter.

4. **For the Armature Current Control Method:** The field current I_f was kept constant (field rheostat fixed) and the armature rheostat R_{AR} was varied in steps. For each step, the armature current I_a and the corresponding motor speed N (from the tachogenerator) were recorded.
5. **For the Field Current Control Method:** The armature voltage was kept constant (armature rheostat fixed) and the field rheostat R_F was varied in steps. For each step, the field current I_f and the corresponding motor speed N were recorded.
6. All readings were tabulated and the corresponding speed-control characteristic curves were plotted.
7. After completing the observations, the rheostats were returned to their maximum safe positions and the supply was switched OFF.

Data Table

Table 2: Readings for Field Control Method

SI. No.	Field Current, I_f (A)	Speed, N (rpm)
1	0.100	1000
2	0.095	1050
3	0.090	1100
4	0.085	1160
5	0.080	1230
6	0.075	1310
7	0.070	1400
8	0.065	1550
9	0.060	1700
10	0.055	1850
11	0.050	2000

Table 3: Readings for Armature Control Method

SI. No.	Armature Current, I_a (A)	Speed, N (rpm)
1	0.200	1000
2	0.220	1100
3	0.240	1200
4	0.260	1300
5	0.280	1450
6	0.290	1550
7	0.300	1650
8	0.310	1780
9	0.315	1850
10	0.320	2000

Graph

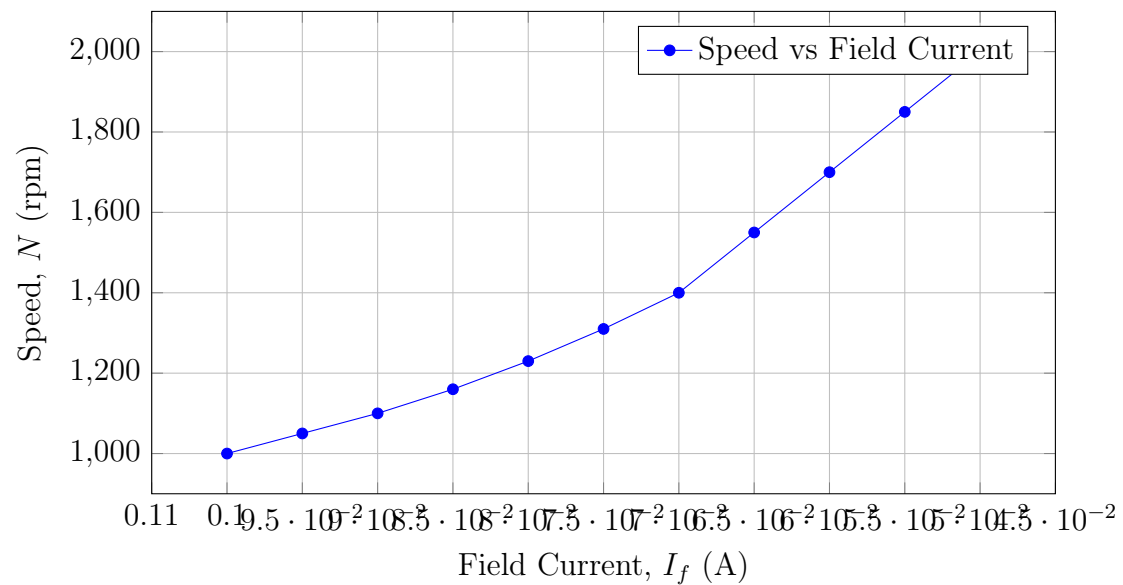


Figure 6: Speed vs. Field Current curve for the Field Control Method

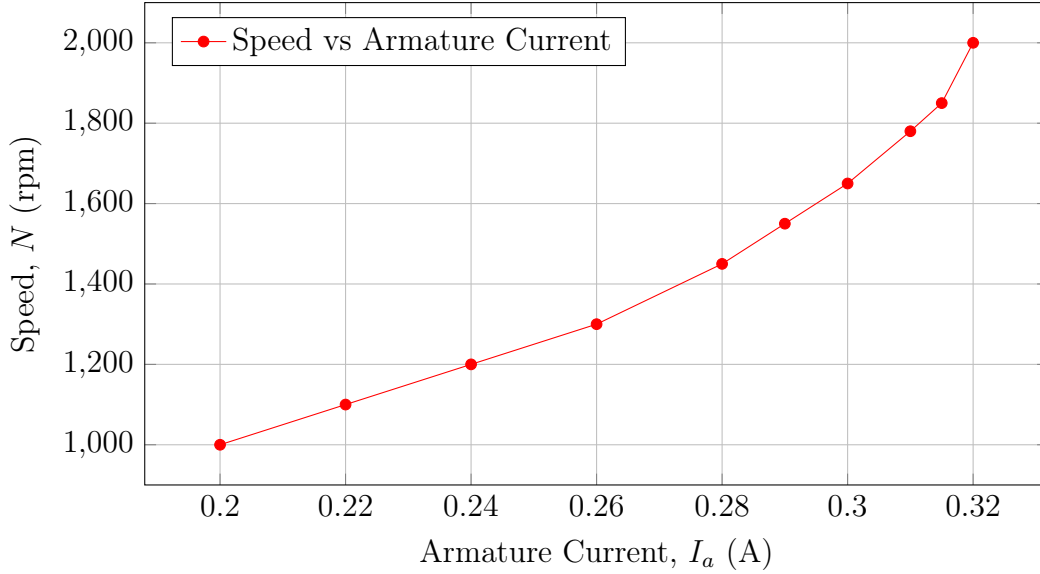


Figure 7: Speed vs. Armature Current curve for the Armature Control Method

Sample Calculation

Percentage change in speed for the Field Control Method (from SI. No. 1 to SI. No. 11):

$$\% \Delta N = \frac{N_{11} - N_1}{N_1} \times 100\% = \frac{2000 - 1000}{1000} \times 100\% = 100\% \quad (4)$$

Percentage change in speed for the Armature Control Method (from SI. No. 1 to SI. No. 10):

$$\% \Delta N = \frac{N_{10} - N_1}{N_1} \times 100\% = \frac{2000 - 1000}{1000} \times 100\% = 100\% \quad (5)$$

Discussion

From Table 2 and Figure 6, it is observed that as the field current I_f decreases (by increasing the field rheostat resistance), the flux ϕ decreases and the speed N increases. This confirms the inverse relationship $N \propto 1/\phi$ predicted by theory. A relatively small change in field current produced a large change in speed, verifying that the field control method gives a wide range of speed control above the rated speed with low power loss, since the field circuit carries only a small current.

From Table 3 and Figure 7, it is observed that as the armature current I_a increases (by decreasing the armature rheostat resistance), the voltage available across the armature increases, which increases the back EMF and hence the speed. The speed varies almost linearly with armature current in this range, consistent with the equation $N \propto (V - I_a R_a)$. However, this method is less efficient than field control since the series rheostat dissipates a considerable amount of power as heat, and it can only be used to control speed below

or up to the rated value.

Minor deviations of the experimental points from the ideal straight-line/inverse relationship can be attributed to armature reaction, contact resistance in the rheostats, meter reading errors, and slight fluctuations of the supply voltage during the experiment.

Precautions

1. The field rheostat R_F must be kept at minimum resistance and the armature rheostat R_{AR} at maximum resistance before starting the motor, to limit the starting current and avoid loss of field excitation.
2. The motor must always be started using the 3-point starter, never by direct switching, to prevent excessive starting current.
3. Readings should be taken only after the speed has stabilized at each setting.
4. All meter connections (polarity and range) should be checked carefully before switching on the supply.
5. The field circuit of a DC shunt motor should never be opened while the armature is energized, as the loss of field current could cause dangerously high speed (motor “running away”).
6. Loose connections should be avoided, and all terminals should be tightened properly.
7. The supply should be switched off immediately if any abnormal noise, sparking, or overheating is observed.

Conclusion

The speed control characteristics of a DC shunt motor were successfully observed using both the Armature Current Control Method and the Field Current Control Method. The experimental results confirmed the theoretical relationship $N \propto (V - I_a R_a)/\phi$. It was verified that the field control method is more efficient and suitable for speeds above the rated value, while the armature control method is suitable for speeds below the rated value but is comparatively less efficient. The experiment provided practical insight into how DC shunt motor speed can be varied for different industrial applications.

Bibliography

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